

Problem Set: Vectors, Trigonometry, and Frames of Reference — Solutions

1. **Force components** ($\vec{F} = 50 \text{ N}$ at 37° above horizontal).

(a) $F_x = F \cos \theta = (50 \text{ N}) \cos 37^\circ \approx (50)(0.799) \approx \boxed{40 \text{ N}}$

(b) $F_y = F \sin \theta = (50 \text{ N}) \sin 37^\circ \approx (50)(0.602) \approx \boxed{30 \text{ N}}$

Check: $\sqrt{40^2 + 30^2} = \sqrt{1600 + 900} = \sqrt{2500} = 50 \text{ N}. \checkmark$

2. **Magnitude and direction** ($v_x = 12 \text{ m/s}$, $v_y = 5 \text{ m/s}$).

(a) $|\vec{v}| = \sqrt{v_x^2 + v_y^2} = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = \boxed{13 \text{ m/s}}$

(b) $\theta = \arctan\left(\frac{v_y}{v_x}\right) = \arctan\left(\frac{5}{12}\right) \approx 22.62^\circ \approx \boxed{23^\circ}$ north of east.

3. **Frame of reference — train.**

- (a) Take east as positive. Passenger moves at $+1.5 \text{ m/s}$ relative to the train, which moves at $+25 \text{ m/s}$ relative to the ground.

$$v_{\text{passenger/ground}} = 1.5 + 25 = \boxed{26.5 \text{ m/s east}}$$

- (b) The car moves at -15 m/s (west) relative to the ground.

$$v_{\text{passenger/car}} = v_{\text{passenger/ground}} - v_{\text{car/ground}} = 26.5 - (-15) = \boxed{41.5 \text{ m/s east}}$$

4. Frame of reference — boat in river.

Boat velocity relative to water: $(0, 4.0) \text{ m/s}$ (north).

Water velocity relative to ground: $(3.0, 0) \text{ m/s}$ (east).

Boat velocity relative to ground:

$$\vec{v}_{\text{boat/ground}} = (3.0, 4.0) \text{ m/s}.$$

$$(a) \quad |\vec{v}| = \sqrt{3.0^2 + 4.0^2} = \sqrt{9 + 16} = \sqrt{25} = \boxed{5.0 \text{ m/s}}$$

(b) Measuring east of north, the east component is opposite the angle and the north component is adjacent:

$$\theta = \arctan\left(\frac{3.0}{4.0}\right) \approx 36.87^\circ \approx \boxed{37^\circ} \text{ east of north.}$$

5. Energy unit conversions ($E = 1.0 \times 10^6 \text{ J}$).

(a) Kilocalories:

$$E = 1.0 \times 10^6 \text{ J} \times \frac{1 \text{ kcal}}{4184 \text{ J}} \approx \boxed{239 \text{ kcal}}$$

(About 240 “Calories” on a food label.)

(b) Kilowatt-hours:

$$E = 1.0 \times 10^6 \text{ J} \times \frac{1 \text{ kWh}}{3.6 \times 10^6 \text{ J}} \approx \boxed{0.28 \text{ kWh}}$$

(c) Electron-volts:

$$E = 1.0 \times 10^6 \text{ J} \times \frac{1 \text{ eV}}{1.6 \times 10^{-19} \text{ J}} = \boxed{6.25 \times 10^{24} \text{ eV}}$$